PRACTICE EXAM 16 / EXAM SET 03

Domain 2: Anatomy and Physiology

8 scored questions	4 answer choices per question	Answer key + explanations included

Instructions

Choose the single best answer (A, B, C, or D) for each question. Mark your responses before checking the answer key and detailed explanations at the end of this document. These practice questions are not official NHA examination items.

QUESTIONS

Q01 | Body Structures and Organ Systems

> **SCENARIO** A patient with provider-diagnosed sinus node dysfunction is scheduled for pacemaker placement. The patient asks whether the device will pump blood instead of the heart muscle. The medical assistant reinforces the provider's explanation of its electrical role.

Which normal function is supported when a pacemaker supplies an impulse because the sinus node fires too slowly?

A. Moving blood out of the left ventricle by replacing its muscle contraction.
B. Initiating an electrical signal that normally begins in the sinoatrial node.
C. Opening the aortic valve by replacing its supporting connective tissue.
D. Adding oxygen to blood by replacing gas exchange in the lungs.

Q02 | Pathophysiology and Disease Processes

> **SCENARIO** An older adult has provider-diagnosed cataracts and is reviewing the planned operation with the medical assistant. The patient says, "I thought the cloudy part was the retina." The assistant uses the surgeon's teaching sheet to clarify the disease process.

Which explanation correctly describes the change responsible for cataract-related visual blurring?

A. The retina separates from the supporting tissue at the back of the eye.
B. Pressure damages the optic nerve and reduces transmission of visual signals.
C. The cornea develops an irregular shape that changes how incoming light bends.
D. The lens becomes cloudy and interferes with light reaching the retina clearly.

Q03 | Body Structures and Organ Systems

SCENARIO A clinic's heat-safety lesson compares two people doing the same activity at the same air temperature, with similar hydration and clothing. One is in dry air; the other is in very humid air and remains covered with sweat. The lesson describes the brain activating sweat glands as body temperature rises.

Which explanation best describes why sweating may cool the person in humid air less effectively?

A. Less sweat evaporates from the skin, so less body heat is removed by evaporation.
B. More sweat evaporates from the skin, so heat moves from the air into the body.
C. Sweat glands stop receiving nerve signals whenever moisture remains on the skin.
D. Sweat cools the body mainly by increasing the temperature of circulating blood.

Q04 | Body Structures and Organ Systems

SCENARIO A patient being evaluated by the provider for coughing during meals is reviewing a swallowing diagram. The medical assistant points to the flap that normally folds over the laryngeal entrance during swallowing to help keep food out of the airway.

Which structure is shown?

A. Uvula
B. Vocal fold
C. Epiglottis
D. Soft palate

Q05 | Pathophysiology and Disease Processes

SCENARIO A patient with provider-diagnosed lactose intolerance says, "Does this mean my immune system attacks the protein in milk?" The medical assistant reinforces the provider's explanation of why milk can cause bloating and diarrhea.

Which response correctly describes lactose intolerance?

A. It results from an immune reaction to the proteins found in milk.
B. It results from too little lactase to digest the sugar found in milk.
C. It results from too little bile to digest the fat found in milk.
D. It results from an obstruction that prevents milk from leaving the stomach.

Q06 | Pathophysiology and Disease Processes

> **SCENARIO** A patient with documented coronary artery disease and stable exertional angina has no symptoms at today's visit. The provider has ordered an exercise stress test and explained that it will help evaluate the heart under increased workload. The medical assistant reinforces that explanation.

Why can exercise reveal a limitation that is less apparent at rest?

A. Heart muscle needs more oxygen during exercise, and narrowed coronary arteries may limit the needed increase in blood supply.

B. The heart obtains its oxygen directly from blood inside its chambers, and exercise stops that direct transfer.

C. The coronary arteries normally close during exercise, and the test measures how long they remain closed.

D. Heart muscle uses less oxygen during exercise, and narrowed coronary arteries cause an excessive oxygen supply.

Q07 | Body Structures and Organ Systems

> **SCENARIO** A patient undergoing an infertility evaluation has a provider order to check whether the fallopian tubes are open. The patient asks why open tubes matter when the ovaries are already releasing eggs. The medical assistant reinforces the provider's anatomy explanation.

Which normal role of the fallopian tubes best explains the purpose of this evaluation?

A. They produce the eggs that are released during each menstrual cycle.

B. They provide the usual site where sperm can meet an egg and a passage toward the uterus.

C. They provide the uterine lining where a developing embryo normally implants.

D. They form the lower uterine opening through which menstrual blood enters the vagina.

Q08 | Pathophysiology and Disease Processes

In a person without conditions that alter red blood cell survival or interfere with the assay, what does a hemoglobin A1C test primarily reflect?

A. The blood glucose concentration at the exact moment the sample is collected.

B. The amount of insulin produced during the meal immediately before testing.

C. Average blood glucose exposure over approximately the previous 3 months.

D. The amount of glucose excreted into urine during the previous 24 hours.

ANSWER KEY & EXPLANATIONS

Practice Exam 16 | Domain 2: Anatomy and Physiology | 8 scored questions

01 B	02 D	03 A	04 C	05 B	06 A
07 B	08 C				

DETAILED EXPLANATIONS

Q01 | Correct answer: B

Initiating an electrical signal that normally begins in the sinoatrial node.

The sinoatrial node normally initiates the electrical signal that starts a heartbeat. When indicated and programmed to do so, a pacemaker can supply an electrical impulse when the natural rhythm is too slow. The device stimulates the heart; it does not itself replace the pumping action of the ventricular muscle.

Q02 | Correct answer: D

The lens becomes cloudy and interferes with light reaching the retina clearly.

A cataract is a cloudy area in the lens, which normally helps focus light onto the retina. Lens clouding can make vision hazy or blurred. Retinal detachment and optic nerve damage involve other structures and do not describe the cataract process documented in this patient.

Q03 | Correct answer: A

Less sweat evaporates from the skin, so less body heat is removed by evaporation.

Evaporation of sweat removes heat from the skin. High humidity reduces evaporation, so visible sweating does not necessarily indicate effective cooling. This response links nervous-system control with the skin's role in maintaining temperature; sweat that remains on the skin provides less evaporative cooling.

Q04 | Correct answer: C

Epiglottis

The epiglottis is a flap of cartilage that folds over the entrance to the larynx during swallowing as part of airway protection. The soft palate and uvula help separate the nasal passage from the oral swallowing pathway. Identifying this structure does not establish the cause of the patient's coughing, which requires provider evaluation.

Q05 | Correct answer: B

It results from too little lactase to digest the sugar found in milk.

Lactose intolerance results when insufficient lactase leaves some lactose, the sugar in milk, undigested. Lactase is an enzyme associated with the lining of the small intestine. An immune reaction to milk proteins describes milk allergy, a different mechanism that should not be treated as equivalent to lactose intolerance.

Q06 | Correct answer: A

Heart muscle needs more oxygen during exercise, and narrowed coronary arteries may limit the needed increase in blood supply.

Exercise increases the heart muscle's workload and oxygen requirement. Coronary artery narrowing can restrict the increase in oxygen-rich blood needed to meet this demand, making a supply-demand limitation more evident. The coronary arteries supply the heart muscle; the muscle does not receive its primary oxygen supply directly from blood within the chambers.

Q07 | Correct answer: B

They provide the usual site where sperm can meet an egg and a passage toward the uterus.

Fertilization normally occurs in a fallopian tube, and the tube provides a pathway toward the uterus. A blockage can interfere with sperm reaching the egg or with transport toward the uterus even when ovulation occurs. The ovaries release eggs, while implantation normally occurs in the uterine lining.

Q08 | Correct answer: C

Average blood glucose exposure over approximately the previous 3 months.

The A1C test measures the proportion of hemoglobin with attached glucose and reflects average blood glucose exposure over approximately the preceding 3 months. It does not measure insulin output or urinary glucose. A blood glucose test at a single time point provides different information from this longer-term marker.

End of Practice Exam